

Calling attention to opponents of climate action in climate and health messaging

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Previous research suggests that providing information about the health effects of climate change and the health benefits of climate action can increase public engagement with the issue. We sought to extend those findings with an experiment to test the motivational value of calling attention to opponents of climate action. In February, 2022, we conducted a survey experiment with adults from the USA, quota-sampled to represent the USA population (n=2201). Participants were randomly assigned to a no-message control condition, or one of four message conditions identified as authored by concerned health professionals. These messages warned recipients about the negative effects of climate change on health, and either made no mention of an opponent to climate action, or were messages augmented by identifying one of three opponents: (1) fossil fuel chief executive officers and their lobbyists, (2) politicians, or (3) a combination of the two. Portrayal of opponents to climate action increased attitudinal engagement, support for mitigation policies, and intentions to advocate for climate solutions, compared with message conditions not identifying an opponent—with the combined opponent portrayal tending to result in the largest effects; these effects were evident with audiences across political lines, especially political conservatives. Climate and health messages—with or without portrayal of an opponent—also increased trust in the messengers relative to the no-message control. These findings suggest that identifying opponents to climate action can be advantageous to building support for such action, reducing political issue polarisation, and fostering greater trust in health professionals as climate messengers.

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Introduction

Climate change is one of the greatest threats to human health in the 21st century.¹ The damage is already widespread and severe, harming people's health and safety across the globe. Policies that can reduce carbon emissions and mitigate the effects of climate change also have many benefits to human health, which are direct, immediate, and independent of any effect they have on slowing climate change.¹ For example, multiple studies conclude that the closure of coal-fired power plants produces near-term health benefits via improvements to ambient air quality.^{2,3} Encouraging and empowering the public to collectively engage in advocacy actions—such as contacting their political representatives, petitioning, protesting, and joining and volunteering for climate organisations—is an important way to build the political will for such policies to be implemented.^{4,5}

Presenting information about the human health relevance of climate change has been shown to increase public engagement by fostering feelings of hope regarding society's ability to address the problem,⁶ increasing levels of worry and personal risk perception regarding climate change,⁷ strengthening support for climate policy,⁸ and increasing motivation to take climate advocacy actions.⁹ Moreover, some of these positive effects have been shown to persist among political moderates and moderate conservatives for several weeks,⁷ which is a promising result considering how contentious and politicised the issue has become in the USA.¹⁰ As the most trusted source of information on climate change and health,¹¹ health professionals are well positioned to deliver these messages to the public and policy makers. Additionally, large numbers of health professionals are concerned about the negative effects of

climate change on the health of their patients and community, and are willing to advocate for climate and health solutions.^{12–14}

With this evidence in mind, we explored the potential to strengthen the effectiveness of climate and health messages by developing and testing a novel element: identifying opponents to climate actions. We believed this information would be a beneficial addition for two reasons. First, the Social Identity Model of Collective Action (SIMCA)¹⁵ proposes three predictors of collective political actions: perceived injustice, collective efficacy, and social identity. Perceived injustice refers to people's belief that their group (or another group with which they identify) is being unjustly deprived of benefits or unjustly subjected to grievances. Collective efficacy refers to the belief that collective actions taken by one's group will be effective in achieving the group's goals and addressing the injustice. Social identity refers to one's identification with a social group and with social identity, people can start to perceive the group's injustice and collective efficacy as their own. Together, these three factors predict people's engagement in collective actions; ie, people are more likely to participate in collective actions when they perceive a greater injustice against the group, hold stronger efficacy beliefs regarding the group's collective efficacy, and identify more strongly with the group. In the present study, we focus on the role of perceived injustice, holding information about collective efficacy, and group identity constant across message conditions. Adding a portrayal of powerful, self-interested opponents to climate actions could theoretically increase perceived injustice, which, according to SIMCA, will lead to stronger message effects on collective actions. Similar to how discussing climate change as a human health issue

can make the topic more personally relevant to people,¹⁶ we suspect that identifying opponents to climate action can humanise the issue and provide a more concrete sense of who is responsible for insufficient societal action.

Second, there exists a long-standing climate injustice that, if properly portrayed, is likely to resonate with the American public. Climate denial in the USA is the product of a top-down, well orchestrated counter-movement to manufacture and perpetuate misinformation regarding the reality and consequences of climate change; sow distrust in scientific evidence, climate scientists, and the institution of climate science; and create uncertainty among the public.^{17–19} Exposure to this type of misinformation has been shown to increase the levels of uncertainty surrounding climate change risks.²⁰ The actors behind this climate change counter-movement include the fossil fuel industry and other major corporations, conservative foundations and think tanks, right-wing media personalities, and a substantial segment of conservative politicians.^{21,22} Together, they represent “powerful vectors of misinformation”²⁰ and are a fitting group to be portrayed as opponents of climate actions.

Furthermore, many Americans already see the fossil fuel industry as culpable. In 2021, 60% of American adults viewed the fossil fuel industry to be completely or mostly responsible for causing climate change, and 45% agreed that these companies lie about their contribution to climate change and about its existence in general.²³ Thus, based on the SIMCA, the evidence implicating the fossil fuel industry in the climate change counter-movement, and public perceptions of the culpability of the fossil fuel industry, we have reasons to believe that identifying the industry (specifically fossil fuel chief executive officers [CEOs] and their lobbyists) as opposing actions to protect climate and health will enhance the message effects on collective actions.

More specifically, we propose that a climate and health message with portrayal of an opponent to climate actions will be more effective than the same message without such portrayal (H1a) and a no-message control condition (H1b). To test message effectiveness, we measured three types of outcomes: first, the immediate outcomes of perceived injustice, including anger and perceived responsibility; second, other outcomes relevant to advocacy and collective actions, including mitigation policy support, desire for societal actions, motivation to take actions, and perceived urgency of addressing climate change; and third, the main outcomes of interest, including advocacy intention and behaviour.

In addition to assessing the effect of identifying the fossil fuel industry as opponents to climate action, we also evaluated politicians as opponents. Politicians were selected because, among the key opponents to climate action identified by Dunlap and Brulle,²¹ they are visible and salient to the public. We wanted to establish (RQ1) whether the portrayal of fossil fuel CEOs and lobbyists is

more or less effective than the portrayal of politicians as climate action opponents.

Furthermore, we included a condition combining the portrayal of both fossil fuel CEOs and their lobbyists with politicians as opponents. The oil and gas industry has a well documented history of making financial campaign contributions to legislators with anti-environmental voting records,²⁴ presumably to secure influence over the policy-making process. Although many Americans might not be fully aware of these details, survey evidence suggests this kind of information would provoke a sense of injustice among many voters. For example, few American voters (16%) say they feel like they have a fair say in how the US Government responds to global warming,²⁵ and Americans in both political parties cite the influence of special interest money on elected officials as the top problem with the federal government.²⁶ Thus, we propose that explicitly making the link between the fossil fuel industry and its influence on politicians (H2) will outperform messages portraying either opponent in isolation.

As climate and health messages have been shown to be effective with a broad audience, including people across the political spectrum,^{7,9,27} we wanted to examine whether adding a portrayal of an opponent to climate action makes these messages more or less polarising. Therefore, RQ2 asks whether the effects of opponent portrayal are influenced by participants’ (1) party affiliation and (2) political ideology.

Finally, the current study aims to understand the effect of these portrayals on trust in the message source—health professionals in this case. Calling attention to opponents of action might be perceived by the audience to be controversial, thus leading to perceptions that the message source is less trustworthy. Health professionals would be right to be concerned about this potential lack of trust because they have earned their trust and should protect it, lest their patients and the public might be less likely to heed their health advice. Therefore, we asked whether adding a portrayal of an opponent to climate actions to a climate and health message (RQ3a) reduces trust in health professionals as a source of information about climate change; and which opponent portrayal has the greatest effect on trust (RQ3b), if any.

Methods

Sample

3044 adults from the USA were recruited for the study between Feb 16 and Feb 20, 2022, by Climate Nexus Polling, which used several market research panels in the USA and recruited respondents identified by stratified sampling methods. All panel vendors recruited individual respondents into their panel by various methods (cold calls, in-person recruitment at shopping centres and events, email, online advertisements, etc). Climate Nexus Polling directly sampled respondents

from these panels and collaborated with all partners to deduplicate and uniquely identify every respondent to eliminate overlap that might occur across panels. Climate Nexus Polling used fixed-stratum proportions of the population, set quotas, and identified population parameters for sampling from their certified vendors. Respondents that met the specific quota requirements became part of the target population and a select random set of respondents were contacted (via email or application notification) to take the survey. Contact and recruitment for the survey was directly managed by their panel vendors. Quotas were set to match American census parameters on age, gender, race, ethnicity, education, annual household income, and political party affiliation to approximate a nationally representative sample. All participants were compensated directly by panel providers with either money or a point reward system. The compensation amount was variable depending on the panel provider and participant characteristics. For example, groups that were harder to contact (eg, individuals aged 18–24 years) usually received higher compensation to incentivise their participation. The consent form explained that the purpose of the survey was to learn their opinions about climate change and health.

53 participants declined to consent, 530 failed an attention check measure in the survey and were filtered out accordingly, and 211 did not complete the study. Among those who completed the study, 49 participants took less than half of the median response time or more than 6-times the median response time to complete the study, suggesting a lack of attention. These responses were removed, resulting in a final sample of 2201 participants. A power analysis conducted in G*Power shows that this sample size yields 80% power to detect an effect size (f) of 0.07 (assuming an α level of 0.05, numerator [df] of 4), which is considered a small effect size in communication research.²⁸

Design and protocol

The institutional review board at George Mason University approved the study. An online between-subjects experimental design was used. After giving consent, participants were randomly assigned to one of five conditions: a control condition in which participants were not shown any message (no-message control condition; $n=448$), a second control condition in which participants read a climate and health message without a portrayal of a climate actions opponent (standard message control condition; $n=426$), or one of the three opponent conditions. One condition portrayed “fossil fuel CEOs and their lobbyists” as the opponent (fossil fuels condition; $n=438$), one portrayed “some politicians” (politicians condition; $n=457$), and the last portrayed “fossil fuel CEOs and their lobbyists, and the politicians who are in their pockets” (combined condition; $n=432$). In the final two conditions, we intentionally omitted any explicit mention of the

political party membership of the politicians opposed to climate action because we felt that numerous health professionals would be uncomfortable with an openly partisan message and that such an approach would be polarising among audiences.

All messages included information about the health harms caused by climate change, followed by information about the health benefits of climate solutions. In the three opponent conditions, the opponent and their actions to hide the truth about climate change were explicitly mentioned. The text was identical across the three conditions; only the opponent changed. This opponent paragraph was omitted from the standard message control condition. All messages ended with a call to action, which also included language to foster group efficacy. A sense of collective identity was cultivated throughout the message with the use of the words *we* and *our*, such as “we all want the same things”, “our communities”, and “our efforts to build a better future”. All messages were preceded by the statement, “The following statement was written by a group of concerned health professionals.” The full text of all message conditions is in the appendix (pp 5–8).

See Online for appendix

After message exposure, participants were presented with the post-test questionnaire (full questions are listed in the appendix pp 9–38). Those in the no-message control condition proceeded directly to the post-test after giving their consent. The dependent variables were assessed in the order in which they are presented below.

Measures

Anger was measured by asking participants to use a slider (online questionnaire) to indicate how much anger they were currently feeling about climate change. Other emotions, such as fear, hope, and guilt, were also included to avoid sensitising participants. All emotions were measured on a scale of 1 (not at all) to 100 (a great deal; mean 43.42 [SD 35.07]).

The perceived responsibility of the opponent was measured on a 5-point scale, from 1 (not at all responsible) to 5 (extremely responsible). Participants were asked how responsible, if at all, they thought each of the opponents (fossil fuel CEOs and their lobbyists and politicians who stand in the way of efforts to address climate change) was for climate change (mean_{fossil fuels} 3.76 [SD 1.26]; mean_{politicians} 3.71 [SD 1.31]).

Mitigation policy support was measured by asking participants to indicate how much they support or oppose the four policies (eg, “Transition the U.S. economy from fossil fuels to 100% clean energy by 2050”). Participants were asked to respond on a scale of 1 (strongly oppose) to 5 (strongly support; mean 3.69 [SD 1.15]; Cronbach’s $\alpha=0.91$).

Desire for societal action was measured by assessing whether participants thought the following institutions should be doing more or less to address climate change: the President, US Congress, their state government,

	Participants (n=2201)	Percentage
Gender		
Female	1166	53.0%
Male	1005	45.7%
I identify in a different way	30	1.4%
Age, years		
18–24	227	10.3%
25–34	448	20.4%
35–44	417	18.9%
45–54	307	13.9%
55–64	404	18.4%
≥65	398	18.1%
Education		
Did not graduate from high school	75	3.4%
High school graduate or equivalent	541	24.6%
Some college, but no degree	623	28.3%
Associate's degree	204	9.3%
Bachelor's degree	506	23.0%
Master's degree	224	10.2%
Doctorate or professional school degree	28	1.3%
Race or ethnicity		
White, non-Hispanic or Latino	1548	70.3%
Black, non-Hispanic or Latino	242	11.0%
Other race, non-Hispanic or Latino	111	5.0%
Hispanic or Latino	300	13.6%
Political party affiliation		
Republicans (including leaners)	825	37.5%
Democrats (including leaners)	927	42.1%
Independent (excluding leaners)	314	14.3%
No party or not interested	135	6.1%
Leaners refers to individuals who initially identify as independent, but then report in a follow-up question that they lean towards either the Republican or Democratic party.		
Table 1: Sample demographics		

their local government, and corporations and industry. Response options ranged from 1 (much less) to 5 (much more; mean 3.88 [SD 1.06]; Cronbach's $\alpha=0.95$).

Motivation to take actions was measured with three items using a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The question stem reads: "I am motivated to work with others to...", and the three collective actions assessed were: demand climate actions by the government, hold fossil fuel CEOs and lobbyists accountable, and vote out politicians who stand in the way of efforts to address climate change (mean 3.63 [SD 1.19]; Cronbach's $\alpha=0.90$).

Perceived urgency of addressing climate change was measured with two items. Participants were asked to indicate how much they agreed or disagreed with the following statements on a 5-point Likert scale: "Immediate action is needed to address climate change", and "The longer we wait to take serious action to address climate change, the worse off we will all be" (mean 3.92 [SD 1.25]).

Advocacy behavioural intention was an average of four 7-point items that asked participants to report the likelihood, ranging from 1 (extremely unlikely) to 7 (extremely likely), that they would engage in each of the following behaviours to advocate for climate actions in the next 12 months: contact a government official, vote, donate money, and volunteer their time (mean 3.82 [SD 1.70]; Cronbach's $\alpha=0.89$).

Advocacy behaviour was measured as a dichotomous outcome: clicking or not on an embedded hyperlink to a real external website where participants could take an advocacy action. Specifically, they were asked if they were interested in sending a form email to their Senators asking them to end tax breaks for fossil fuels corporations. A link was provided, and the survey platform automatically recorded whether a participant clicked on the link or not. On average, the click rate was 9.5%.

Trust in health professionals asked participants how much, if at all, they trust health professionals as a source of information about climate change. Response options ranged from 1 (not at all) to 5 (a great deal; mean 3.20 [SD 1.15]).

Analysis

To test H1, for each of the outcomes except advocacy action, an ANOVA model was performed with the comparison of interest. The advocacy action outcome was tested with crosstab analysis with χ^2 test for significance. To test H1a, the standard message control condition (coded as 0) was compared with all three opponent message conditions (coded as 1); the no-message control condition was excluded. To test H1b, the no-message control condition (coded as 0) was compared with the three opponent message conditions (coded as 1); the standard message control condition was excluded.

The same ANOVA and crosstab models were run to test H2 and answer RQ1. All five conditions were included, and the mean of each conditions was compared against one another. We present our analyses later without multiple-comparison correction because such adjustments are applied arbitrarily and inconsistently, can increase type II error, and are no substitute for replication.^{29,30} However, we include Bonferroni-adjusted mean comparisons to account for multiple testing (appendix pp 2–3).

To examine RQ2, simple moderation analyses were run with the PROCESS macro.³¹ The experimental conditions served as the categorical predictor. In one set of models, party identification served as a continuous moderator; in a separate set of models, political ideology was the continuous moderator (appendix p 4). Both sets of models examined all outcome variables.

To answer RQ3, an ANOVA model with all five conditions was performed. Less than 3% of data were missing on all variables and any cases with missing data were excluded listwise in the analysis.

Results

The demographics of the sample are reported in table 1.

H1: the effects of opponent portrayals compared with control conditions

H1 predicts that including an opponent to climate actions will increase the effectiveness of climate and health messages compared with (1) a message with no opponent portrayal, and (2) a control condition with no-message exposure. Message effectiveness was indicated by perceived injustice, including anger and perceived responsibility of the opponent; other important outcomes, including policy support, desire for societal action, motivation, and perceived urgency; and the main advocacy outcomes, including behavioural intention and advocacy action.

We found support for H1a on all outcomes except for advocacy action ($\chi^2=0.81$, $p=0.369$). We also found support for H1b on all outcomes except perceived urgency and advocacy action ($\chi^2=1.09$, $p=0.297$). Thus, H1a and H1b were partly supported, as shown by higher levels of perceived injustice, policy support, desire for societal action, motivation, and behavioural intention in the opponent conditions. Table 2 provides the p values and F statistics for all the aforementioned ANOVA models.

H2 and RQ1: comparing the effects between the three opponent portrayals

H2 and RQ1 are concerned with the relative effectiveness of the three opponent messages. In H2, we predicted that the combined condition (portraying both fossil fuel CEOs and lobbyists and politicians) would be more effective than the two control conditions and the other two single-opponent conditions. In RQ1, we asked if the two single-opponent conditions differed in terms of message effectiveness.

The comparisons between the fossil fuel opponent and the politician opponent conditions serve to answer RQ1; no significant differences were observed between the two conditions (table 3). However, compared with the control conditions, we found a significant effect across a greater number of outcomes in the fossil fuel condition compared with the politician condition. Again, no between-group differences were detected for the advocacy action outcome. Thus, the results show that when portrayed separately, the two types of opponents—fossil fuel CEOs and lobbyists, and politicians—influenced the outcomes in a mostly similar pattern.

The H2 test was made by comparing combined fossil fuel and politician opponent condition and the individual fossil fuel opponent and politician conditions; the combined condition resulted in significantly higher perceived responsibility than either of the two single-opponent conditions (table 3). The combined condition also resulted in higher perceived urgency than the politician condition, but not the fossil fuel condition. We did not observe any other significant differences between

	Opponent portrayal vs standard message condition		Opponent portrayal vs no-message control	
	F	p value	F	p value
Anger	10.38	<0.0013	10.04	0.0016
Perceived responsibility, fossil fuels	4.37	0.037	7.75	0.0054
Perceived responsibility, politicians	6.27	0.012	5.83	0.016
Policy support	6.11	0.014	3.86	0.050
Perceived need for societal actions	7.03	0.0081	6.22	0.013
Motivation to take actions	7.10	0.0078	7.25	0.0072
Perceived urgency of addressing climate change	5.36	0.021	1.55	0.21
Advocacy behavioural intention	4.85	0.028	6.62	0.010

Table 2: Response scores in portrayal of a climate opponent condition compared with standard message and no-message control condition

the combined condition and either of the two single-opponent conditions for the other outcome variables. Therefore, we found partial support for H2.

RQ2: moderating the effects of party identification and political ideology

RQ2 explores the potential moderating effects of party identification and political ideology on the relationship between opponent portrayals and the measured outcomes. No significant moderating effects were observed for party identification; the opponent portrayals exerted similar influences on Democrats, Independents, and Republicans (appendix p 4). Political ideology, however, moderated the effects of the message condition on several outcome measures, including perceived responsibility of fossil fuel CEOs and lobbyists ($F=2.39$, $p=0.049$) and of politicians ($F=2.44$, $p=0.045$), policy support ($F=2.54$, $p=0.038$), motivation ($F=2.46$, $p=0.043$), and perceived urgency ($F=4.76$, $p<0.001$). Among moderates, reading a message that included the portrayal of an opponent mostly did not change their attitudes (table 4). However, among liberals, invoking politicians as climate opponents reduced the message effect. Conversely, among conservatives, invoking a climate opponent consistently increased policy support regardless of the opponent portrayed. The combined condition performed the best among conservatives, prompting stronger responses on all outcomes compared with the two control conditions. Thus, not only did messages portraying climate opponents not backfire among conservatives, these messages led to stronger positive effects on several key outcomes, demonstrating the depolarising potential of this message frame.

RQ3: the effects of opponent portrayals on trust in health professionals

RQ3 asks (1) whether adding an opponent portrayal would decrease overall trust in health professionals as a source of climate change information, compared with a message without such portrayal or no message,

	No-message control condition	Standard message condition	Fossil fuel opponent condition	Politician opponent condition	Combined fossil fuel and politician opponent condition
Anger (1–100)	39.77 (1.67) ^{ab}	39.43 (1.74) ^a	44.49 (1.69) ^c	44.32 (1.65) ^{bc}	48.95 (1.70) ^c
Perceived responsibility, fossil fuels (1–5)	3.64 (0.06) ^a	3.68 (0.06) ^{ab}	3.83 (0.06) ^{bc}	3.77 (0.06) ^{abc}	3.88 (0.06) ^c
Perceived responsibility, politicians (1–5)	3.61 (0.06) ^a	3.60 (0.06) ^a	3.72 (0.06) ^a	3.73 (0.06) ^a	3.91 (0.06) ^b
Advocacy behavioural intention (1–7)	3.67 (0.08) ^a	3.70 (0.08) ^a	3.96 (0.08) ^b	3.83 (0.08) ^{ab}	3.94 (0.08) ^b
Advocacy action (click rate; 0–100%)	8.5% ^a	8.7% ^a	9.6% ^a	10.7% ^a	10.2% ^a
Policy support (1–5)	3.63 (0.05) ^{ab}	3.59 (0.06) ^a	3.75 (0.06) ^{bc}	3.69 (0.05) ^{abc}	3.81 (0.06) ^c
Perceived need for societal actions (1–5)	3.80 (0.05) ^a	3.78 (0.05) ^a	3.91 (0.05) ^{ab}	3.91 (0.05) ^{ab}	4.00 (0.05) ^b
Motivation to take actions (1–5)	3.53 (0.06) ^a	3.52 (0.06) ^a	3.67 (0.06) ^{ab}	3.65 (0.06) ^{ab}	3.78 (0.06) ^b
Perceived urgency of addressing climate change (1–5)	3.89 (0.06) ^{ab}	3.81 (0.06) ^a	3.98 (0.06) ^{bc}	3.87 (0.06) ^{ab}	4.07 (0.06) ^c
Trust in health professionals (1–5)	2.92 (0.05) ^a	3.20 (0.06) ^b	3.31 (0.06) ^b	3.23 (0.05) ^b	3.35 (0.06) ^b

Data are mean (SE) or %. Mean scores with dissimilar superscripts (a, b, and c) between conditions indicate significant differences at $p < 0.05$. Two different superscripts indicate that the mean is not significantly different from means with either individual superscript, but it would be significantly different from a mean with a superscript different to the two listed.

Table 3: Participant responses compared across no-message, standard message, and opponent message conditions

and (2) which opponent message has the most effect on trust, if any. None of the messages resulted in a decrease in trust. In fact, exposure to any of the four messages increased participants' trust in health professionals compared with the no-message control condition ($F=9.62$, $p<0.001$); there were no significant differences in trust gains between the four messages.

Discussion

Drawing on the SIMCA,¹⁵ we examined the effects of a novel addition to climate and health messaging, namely the portrayal of an opponent to climate action. Our findings suggest that this approach heightens the effect of climate and health messages on emotional and motivational responses. Compared with both control conditions, portrayals of a climate opponent increased perceived injustice—as indicated by feelings of anger and perceived responsibility of the opponent—and the intention to engage in collective advocacy actions. Additionally, invoking an opponent led to positive effects on other important outcomes, including policy support, desire for societal action, motivation, and perceived urgency. Among the three opponent messages, the combined condition that invoked both the fossil fuel CEOs and their lobbyists with the politicians in their pockets messages, tended to produce the largest effects.

However, identifying opponents to climate action did not increase the rates of advocacy action. Although the advocacy action rates in the three opponent conditions (which ranged from 9.6% to 10.7%) were 1 to 2 percentage points higher than those in control conditions (averaging 8.6%), these small differences were not statistically significant.

Furthermore, we found no differential effects among Democrats, Republicans, and Independents, but larger effects among political conservatives as compared with liberals and moderates—although we hypothesise that this might be due partly to a ceiling effect among

liberals. However, backfire or boomerang effects have been observed among conservatives for other messaging approaches.^{32,33} Therefore, the current findings corroborate the depolarising benefit of climate and health messages, as previously shown in studies by Kotcher and colleagues.^{7,9,27} Moreover, these findings are consistent with recent polling, which found that conservative Republicans rate their primary care doctor as one of the most highly trusted sources of information about climate change.²⁵

Finally, we found that these messages, identified as having been written by a group of concerned health professionals, had the additional benefit of increasing readers' trust in health professionals as a source of climate change information. This finding is highly reassuring, considering that many health professionals understandably might have reservations about mentioning these opponents and their actions out of concern that such messages are too controversial and could diminish their credibility. Therefore, health professionals who advocate for climate action can be assured that calling attention to opponents of climate action is likely to be effective, depolarising, and conducive to fostering trust in them as messengers.

Message effectiveness alone is not the only criterion required for successful strategic communication. A large body of research suggests that successful communication campaigns often require sustained message repetition so that dissemination of key messages reaches the desired proportion of the target audience for durable population-level changes in the target attitudes and behaviours of interest.³⁴ Moreover, such efforts should aim to also overcome any restraining forces that work against the goals of the communication campaign, such as misinformation and counter-arguments from opponents to climate action.³⁵ Although our study did not assess trust in the fossil fuel industry or politicians opposed to climate action in response to the opponent messages, we

speculate that this message strategy could encourage scepticism around such actors and help to inoculate audiences against future arguments in opposition to action to address climate change and its health effects.³⁶ For example, past research suggests that perceived conflicts of interest due to association with industry actors can erode trust in messengers.³⁷

From a practical perspective, we suggest that groups of health professionals including the Medical Society Consortium for Climate and Health and the Global Climate and Health Alliance are well positioned to incorporate the insights from this study into their communication efforts in the form of strategically placed op-eds, public presentations, and testimonies.³⁸ Such messaging is likely to be less appropriate in a clinical context where the conversation should primarily be focused on the patient's concerns.³⁹

Although our study was inspired by and produced results consistent with the SIMCA, other theoretical perspectives could help to explain our findings. For example, research on reluctant altruism suggests that individuals might be motivated to engage in some collective actions (eg, blood donation) when they do not trust others to cooperatively engage in those prosocial behaviours.⁴⁰ It is possible that learning about opponents to climate action helped underscore the importance of engaging in collective action to address climate change, with increased advocacy intentions to counter this opposition. Future research could investigate this theoretical mechanism more directly.

Although the focus of this study was not on testing the effectiveness of a climate and health message without portrayal of opponents (ie, the standard message control condition), we note that there were few differences in the outcomes between the standard message and the no-message control condition. To some extent, this challenges previous research that finds exposure to messages about the health implications of climate change increases engagement with the issue.⁷⁻⁹ One possibility identified in previous research is that increasing the saliency of threats to one's personal health can have a paradoxical effect such that it increases people's concern about the issue while also reducing their willingness to engage in advocacy, because they feel less capable of devoting time, money, or attention to advocacy.⁴¹

This study is not without limitations. First, although survey experiments allow for greater experimental control, the results might not be as generalisable as they would be with a field experiment and in a more naturalistic setting. Field experiments allow for greater ecological validity and allow one to assess other criteria beyond message effectiveness, such as how easy it is for communication practitioners to adopt, implement, and maintain the use of a communication strategy over time.^{42,43} Second, the treatments reported here were administered only once and all the measures were

	Liberals (n=702)				Moderates (n=745)				Conservatives (n=754)				Combined fossil fuel and politician opponents		
	Control	Standard	Fossil fuel opponent	Politician opponent	Combined fossil fuel and politician opponents	Control	Standard	Fossil fuel opponent	Politician opponent	Combined fossil fuel and politician opponents	Control	Standard		Fossil fuel opponent	Politician opponent
Perceived responsibility, fossil fuels	4.26 (0.10) ^{ab}	4.41 (0.11) ^b	4.36 (0.10) ^{ab}	4.12 ^a (0.10)	4.30 (0.10) ^{ab}	3.64 (0.10) ^a	3.68 (0.10) ^a	3.89 (0.10) ^a	3.88 (0.10) ^a	3.82 (0.10) ^a	3.06 (0.10) ^a	3.10 (0.10) ^{ab}	3.27 (0.10) ^{abc}	3.34 (0.10) ^{bc}	3.50 (0.10) ^c
Perceived responsibility, politicians	4.27 (0.10) ^a	4.25 (0.11) ^a	4.26 (0.10) ^a	4.08 (0.10) ^a	4.35 (0.10) ^a	3.75 (0.10) ^a	3.66 (0.10) ^a	3.74 (0.10) ^a	3.86 (0.10) ^a	3.92 (0.10) ^a	2.89 (0.10) ^a	3.02 ^{ab} (0.10)	3.19 (0.10) ^{bc}	3.27 (0.10) ^{bc}	3.42 (0.10) ^c
Mitigation policy support	4.35 (0.09) ^{ab}	4.41 ^b (0.09)	4.36 (0.09) ^{ab}	4.15 (0.08) ^a	4.33 (0.08) ^{ab}	3.62 (0.08) ^a	3.70 (0.08) ^{ab}	3.80 (0.08) ^{ab}	3.80 (0.08) ^{ab}	3.91 ^b (0.09)	2.97 (0.08) ^{ab}	2.82 (0.08) ^a	3.11 (0.08) ^b	3.15 (0.08) ^b	3.14 (0.09) ^c
Motivation to take actions	4.21 (0.09) ^a	4.29 (0.10) ^a	4.26 (0.09) ^a	4.08 (0.09) ^a	4.27 (0.09) ^a	3.54 (0.09) ^a	3.55 (0.09) ^a	3.68 (0.09) ^a	3.70 (0.09) ^a	3.76 (0.09) ^a	2.89 (0.09) ^a	2.88 (0.09) ^a	3.09 (0.09) ^{ab}	3.18 (0.09) ^b	3.28 (0.09) ^c
Perceived urgency of addressing climate change	4.71 (0.10) ^b	4.58 ^b (0.10)	4.55 (0.10) ^b	4.23 (0.09) ^a	4.58 (0.09) ^b	3.89 (0.09) ^a	3.95 (0.09) ^a	4.11 (0.09) ^a	3.98 (0.09) ^a	4.15 (0.10) ^a	3.12 (0.09) ^a	3.06 (0.09) ^a	3.29 (0.09) ^{ab}	3.42 (0.09) ^b	3.46 (0.10) ^c

Data are mean (SE). Within ideological subgroups, means with dissimilar superscripts (a, b, and c) between conditions indicate significant differences at p<0.05. Two different superscripts indicates that the mean is not significantly different from other means with either individual superscript, but it would be significantly different from a mean with a different superscript.

Table 4: Participant responses compared across message conditions and political ideology for outcomes with significant interactions

Data are mean (SE). Within ideological subgroups, means with dissimilar superscripts (a, b, and c) between conditions indicate significant differences at $p < 0.05$. Two different superscripts indicates that the mean is not significantly different from other means with either individual superscript, but it would be significantly different from a mean with a different superscript.

Table 4: Participant responses compared across message conditions and political ideology for outcomes with significant interactions

assessed immediately after exposure. Thus, we are unaware of how long the observed effects would last or how these treatments would perform if repeated over time in a competitive media environment. The questionable durability of treatment effects found in survey experiments has long been a concern in social sciences.⁴⁴ We also note that our study used an online non-probability sample of participants. Research has found that such samples generally tend to produce generalisable estimates of experimental treatment effects, indistinguishable from probability-based sampling approaches.⁴⁵ However, as of 2022, around 6% of the American population does not use the internet,⁴⁶ with adoption rates somewhat lower among older adults, and people with lower levels of income and education.⁴⁷ It is possible that this small proportion of non-internet users could be systematically different in some way compared with the internet-users that participated in our study, representing a small source of potential sample bias. Another long-standing concern in social science experiments is a phenomenon referred to as demand effects—ie, that participants might intuit the researcher's hypotheses and behave in a manner consistent with those expectations, potentially biasing study results. However, a recent large-scale study with over 12 000 participants that replicated a variety of experimental designs found that revealing the researcher's intent did not alter treatment effects, and providing financial incentives did not consistently induce demand effects.⁴⁸ Although such evidence calls into question the extent to which demand effects threaten the validity of research, we acknowledge that the influence of demand effects might vary across study contexts and our findings should be interpreted with these issues in mind.

Much research has been published on the actions of the climate change counter-movement,^{17–19} but not many studies have investigated the effect of presenting these facts to the American public and how it influences their willingness to engage in collective actions. We believe this study has provided initial, but compelling, evidence for the effectiveness of this approach in climate change messaging. Calling attention to opponents of climate actions can increase intentions to engage in collective advocacy actions and foster more trust in health professional messengers.

Contributors

JK, KL, and EM conceived and designed the experiment. EM, JK, KL, JC, and RG wrote the stimulus material. JK and KL analysed the data and wrote the first draft. JK revised the manuscript with feedback from all coauthors.

Declaration of interests

JK and EM report grants from the Energy Foundation and WHO during the conduct of this study. All other authors declare no competing interests. The funder had no role in the design, analysis, or interpretation of this research.

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